

Snowy Scenes: A Multimodal Multitask Dataset Toward Snow-Tonomous Vehicles

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Abstract—Robust perception in snowy conditions remains a critical challenge for autonomous vehicles. To drive progress in this area, we introduce *Snowy Scenes*, the first multimodal multitask dataset captured entirely in real-world snowy conditions, combining high-resolution 128-beam LiDAR data with synchronized RGB and thermal images. *Snowy Scenes* includes over 5 K expertly annotated LiDAR scans with 3D bounding boxes, per-point semantic labels, and snow particle annotations. This rich annotation set enables research on 3D object detection, semantic segmentation, and point cloud denoising, both as standalone tasks and within joint multitask learning frameworks. By benchmarking existing methods, our dataset paves the way for the next generation of perception models designed to handle the complexities of snowy winter weather.

Index Terms—Autonomous systems, data collection, sensor fusion, object detection, semantic segmentation, signal denoising.

I. INTRODUCTION

AUTONOMOUS driving systems have made remarkable progress in recent years. However, their performance in adverse weather conditions remains a critical challenge. This limitation stems from a fundamental challenge: adverse winter conditions inherently degrade sensor quality. The relative lack of datasets captured under such conditions is an additional critical factor, constraining the capacity of perception models to mitigate this degradation in real-world winter conditions. For instance, harsh weather, particularly snowfall, introduces unique challenges such as sensor occlusions, noisy readings, and reduced visibility, which can severely degrade the performance

of state-of-the-art perception algorithms. Despite the clear need, the implementation of weather-resistant perception models remains largely unexplored due to a lack of relevant benchmark datasets.

Although there exists a substantial corpus of research [1], [2], [3], [4], [5], [6], [7] introducing various datasets featuring all-weather conditions (see Table I), these studies exhibit several deficiencies in multiple aspects. *First*, most of these datasets have a limited amount of annotated data dedicated to harsh weather conditions. For instance, snow covers only 2% of the entire ZOD [4] dataset and 12% of Ithaca365 [5]. *Second*, while certain datasets focus only on winter conditions, they evaluate only a single perception task, such as 3D object detection, as in the case of the CADC dataset [6], or 3D semantic segmentation, as in SemanticSTF [3]. *Third*, none of these datasets provides ground truth annotations to benchmark diverse multimodal 3D perception tasks in winter conditions. To address these limitations, we introduce *Snowy Scenes*, a multimodal autonomous driving dataset specifically designed to address multiple perception challenges posed by snowy winter conditions. Captured in the Otaniemi district of Espoo, Finland, during winter, the dataset comprises real-world urban driving scenarios featuring falling snowflakes, accumulated snow, and occluded road markings under both daytime and late-evening conditions.

The dataset includes high-quality multimodal sensor readings captured by one 128-beam LiDAR, one 2.3 MP RGB camera, and three 0.3 MP thermal cameras, providing a benchmark for diverse perception models in snowy environments. Different from existing datasets, as illustrated in Fig. 1, *Snowy Scenes* provides not only ground truth 3D bounding boxes but also point-wise annotations of 27 semantic categories, including falling snowflakes, which is crucial to study the detection and removal of snow particles, a.k.a., point cloud denoising or desnowing.

Our extensive benchmark experiments reveal that voxel-based methods are the superior approach when dealing with high-resolution LiDAR under snowy conditions, as they exhibit the highest performance in semantic segmentation and denoising tasks. In contrast, point-based models exhibit low performance across all benchmarking tasks due to their limited receptive fields. Furthermore, our findings indicate that unsupervised point cloud denoising methods are not resolution-agnostic, thereby demonstrating suboptimal performance. These findings clearly show that the currently available advanced perception models are not ready for snowy winter conditions. Therefore, we strongly believe that our new dataset can pave the way toward the development of *snow-autonomous* vehicles. This letter provides three key contributions:

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The dataset and benchmark models are publicly available at: <https://github.com/snowyscenes/dataset>.

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TABLE I
COMPARISON OF SNOWY SCENES AGAINST EXISTING OUTDOOR PUBLIC DATASETS WITH 3D ANNOTATIONS. BB, SL, AND SF REFER TO 3D ANNOTATIONS FOR BOUNDING BOXES, SEMANTIC LABELS, AND SNOW FLAKES, RESPECTIVELY. - REPRESENTS UNSUPPORTED OR NO STATISTICS

Dataset	Year	Modality				3D Annotation			#Frames	#Classes	Snow
		LiDAR	RGB	Thermal	GNSS/IMU	BB	SL	SF			
KITTI [8]	2012	64	90°	-	✓/✓	✓	-	-	15K	8	-
KAIST [9]	2018	32	26°	25°	✓/✓	✓	-	-	95K	3	-
nuScenes [10]	2019	32	360°	-	✓/✓	✓	✓	-	40K	23	-
Waymo [11]	2020	64*	360°	-	✓/✓	✓	✓	-	230K	4	-
A2D2 [12]	2020	16	360°	-	✓/✓	✓	-	-	12K	14	-
Argoverse 2 [13]	2021	32	360°	-	- / -	✓	-	-	150K	30	-
SemanticKITTI [14]	2021	64	-	-	✓/✓	-	✓	-	43K	28	-
DENSE [15]	2019	64	120°	57°	- / -	✓	-	-	12K	5	✓
CADC [6]	2021	32	360°	-	✓/✓	✓	-	-	7K	10	✓
WADS [7]	2021	64	-	-	- / -	-	✓	✓	2K	22	✓
Ithaca365 [5]	2022	128	60°	-	✓/✓	✓	-	-	7K	6	✓
Boreas [16]	2023	128	81°	-	✓/✓	✓	-	-	7K	3	✓
ZOD [4]	2023	128	120°	-	✓/✓	✓	-	-	100K	29	✓
SemanticSTF [3]	2023	64	-	-	- / -	-	✓	-	2K	21	✓
MSU-4S [2]	2024	64	150°	-	✓/✓	✓	-	-	100K	3	✓
MAN TruckScenes [1]	2024	64	360°	-	✓/✓	✓	-	-	30K	27	✓
Snowy Scenes	2026	128	70°	72°**	✓/✓	✓	✓	✓	5K***	27	✓

* Beam numbe for the top spinning LiDAR. ** Combined HFOV of 3 thermal cameras. *** Total amount of collected data is over 22K.

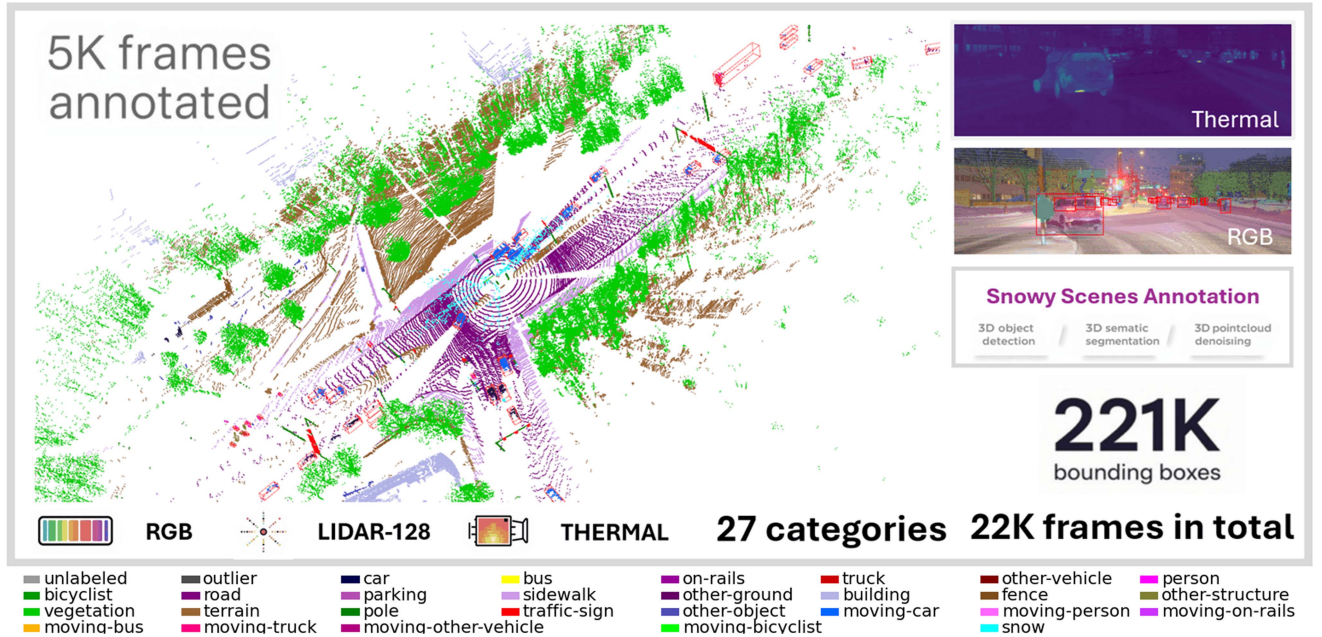


Fig. 1. We introduce Snowy Scenes, a dataset consisting of a high-resolution 128-beam LiDAR point cloud synchronized with RGB and thermal cameras. The dataset includes dense point-wise annotations of 27 semantic categories, including falling snowflakes (depicted in cyan), as well as 3D bounding boxes to benchmark de-snowing, semantic segmentation, and object detection tasks under snowy weather conditions.

- We introduce a new multimodal dataset, named Snowy Scenes, with over 22 K data collected using a high-resolution LiDAR coupled with RGB and thermal cameras in snowy winter conditions only.
- We provide a comprehensive human annotation of over 5 K sequential LiDAR scans, containing 221 K precise bounding boxes for object detection, per-point semantic segmentation, and labeling of falling snow particles. 60% of annotated Snowy Scenes scans has active snowfall.
- Leveraging Snowy Scenes, we benchmark various state-of-the-art models for individual and joint learning of three

key perception tasks: 3D object detection, 3D semantic segmentation, and 3D point cloud denoising. These benchmarking efforts provide comprehensive baselines to support future research on multimodal sensor fusion and multitask learning in snowy weather.

To the best of our knowledge, Snowy Scenes is the first dataset that allows for the learning of multiple 3D perception tasks in a multimodal setup, logged under snowy winter conditions. A devkit with evaluation code and configuration files to reproduce the experiment findings will be made publicly available with the dataset upon publication.

II. RELATED WORK

Autonomous driving has undergone significant advancements over the last decade [8], [9], [10], [11], [12], [13], [14], with notable benchmarks such as KITTI [8], nuScenes [10], and Waymo [11] providing large-scale multimodal datasets for urban driving scenarios. However, these datasets primarily capture clear weather conditions, limiting their applicability in winter weather.

Table I provides an overview of the dataset landscape and compares our proposed Snowy Scenes with the existing body of work in the literature. In this table, we separated the datasets into two groups. The top half encompasses datasets that exclude snow, whereas the bottom half is composed of datasets that explicitly involve snow. It is, however, important to note that in some datasets in this bottom group, snow constitutes a negligible fraction of the whole set. For instance, snow covers only 2% of the entire dataset, i.e., about 2 K frames, in ZOD [4], and 12% (i.e., 800 frames) in Ithaca365 [5]. The snowy data rate goes up to 33.4% in SemanticSTF [3], corresponding to approximately 700 frames. Other datasets, including MSU-4S [2] and MAN TruckScenes [1], have more samples of snowy conditions, but they only provide 3D bounding boxes without semantic labels. In the case of Boreas [16], 3D bounding box annotations are obtained in sunny weather.

Notably, CADC [6] and WADS [7] are the only datasets that focus exclusively on snowy winter conditions, making them the most similar to Snowy Scenes. CADC [6] is the first to provide a fully snowy dataset for 3D object detection, while WADS [7] is the first to provide 3D semantic segmentation as well as the labeling of individual snowflakes. Despite the substantial contributions of both datasets to the research on weather-resistant perception, they exhibit significant shortcomings. CADC [6] suffers from 32-beam low-resolution point cloud data, and WADS [7] is a unimodal dataset. Furthermore, both datasets lack diversity in annotations. In contrast, as shown in Table I, our Snowy Scenes dataset has a diverse sensor suite, including 128-beam high-resolution LiDAR, RGB cameras, thermal imaging, and GNSS/IMU data. In contrast to all other datasets in Table I, Snowy Scenes takes a major leap forward with the most extensive set of annotations, including 3D object bounding boxes, per-point segmentation, and snowflake labeling across the entire 5 K point clouds.

Note that a number of additional datasets are excluded from Table I, despite the presence of snowy conditions. This is because they either lack 3D annotations (e.g., MUSES [17]) or omit LiDAR as in BDD100K [18]. RADIATE [19] is also excluded because it provides only 2D bounding boxes on radar images without 3D annotations.

III. THE SNOWY SCENES DATASET

A. Vehicle and Sensor Setup

As depicted in Fig. 2, our vehicle (Ford Mondeo, 2017) is equipped with a 360-degree, high-resolution 128-beam LiDAR, a front-facing RGB camera, and three thermal cameras. Additionally, the vehicle has a GNSS/IMU system for accurate ego-positioning. Detailed specifications for each sensor can be found in Table II. All sensors are time-synchronized using a GNSS/IMU sensor and calibrated as detailed in Section III-B.



Fig. 2. Sensor setup for the data logging vehicle.

TABLE II
SENSOR SPECIFICATIONS OF SNOWY SCENES

Sensors	Details
LiDAR	1×Velodyne VLS-128 rotating 3D laser scanner, HFOV 360°, VFOV 40°, HRES 0.1° to 0.4°, VRES 0.11°, channels 128, wavelength 903 nm, range up to 200m, and frame rate 10 Hz.
RGB Camera	1×Basler acA1920-155um, RGB front-looking camera, HFOV 70°, VFOV 48°, and resolution 1920×1200 (2.3 MP).
Thermal Camera	3×Teledyne Flir ADK, HFOV 24°, VFOV 19°, and resolution 640×512 (0.3 MP). The combined HFOV 72°, VFOV 19°, and resolution 1920×512 (1.0 MP).
GNSS/IMU	1×Novatel PowerPak7-E1 Inertial Navigation System (INS), 6 axes, L1/L2 RTK with frame rate 200 Hz, 0.01 m position accuracy.

* HFOV/VFOV and HRES/VRES stand for the horizontal/vertical field of view and resolutions, respectively.

B. Sensor Calibration and Synchronization

The **global frame** is an East-North-Up (ENU) frame defined according to the ETRS-TM35FIN coordinate system. It is a Universal Transverse Mercator projection (ETRS89 / UTM zone 35 N) that spans the entire country of Finland. The up coordinate is defined as the ellipsoidal height. The vehicle **ego-frame** and the **LiDAR frame** are the same. Its x -axis is positive to the right, the y -axis is positive forward, and the z -axis is positive upwards. The **camera frame** is located in the camera focal point (x -axis positive right, y -axis down, and z -axis forward), defined with a 4×4 camera extrinsic matrix that maps data from this sensor to the LiDAR frame. The camera intrinsic matrix for the pinhole model is also provided, along with the Brown-Conrady distortion parameters, which account for lens distortion. All thermal cameras also have similar coordinate frames and parameter matrices that transform their data into the LiDAR frame, which is used as the ego-frame.

The time synchronization of sensors is performed using the pulse per second (PPS) signal and NMEA serial messages sent by the Novatel GNSS/IMU to other sensors, which lock their internal clocks to the main time source. RGB and thermal cameras are hardware-triggered by the same Novatel GNSS/IMU sensor at 10 Hz. LiDAR data is logged continuously during data collection, and its measurements are later motion compensated (See Section III-C). For each motion-compensated LiDAR scan, the closest-in-time RGB and thermal images are associated with it. The data-collection frame rate is 10 Hz.

C. LiDAR Motion Compensation

Due to the nature of spinning LiDAR measurements, it is crucial to compensate for the distortion caused by motion during continuous measurements. The motion distortion is

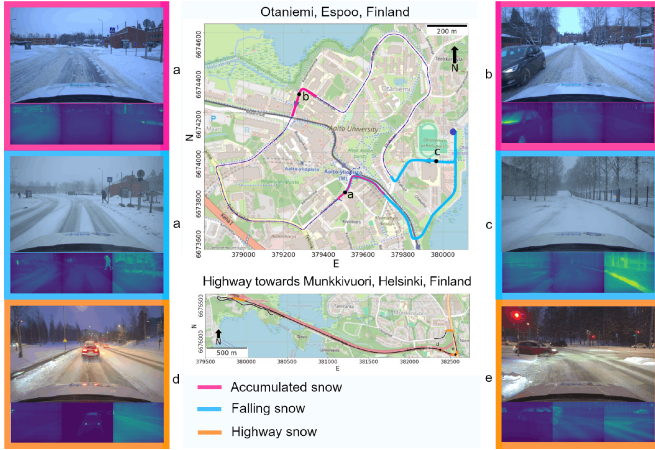


Fig. 3. Route maps with sample RGB and thermal images, taken from five locations: a–e. The two Otaniemi routes align with each other, and the first two images were taken at the same location at different times. The color of each image frame matches its corresponding routes on the map. The thicker colored parts of the routes indicate the annotated frames.

compensated by utilizing a post-processed GNSS/IMU trajectory derived from Novatel Inertial Explorer software and data from nearby GNSS base stations. First, the LiDAR trajectory, defined as the pose of the LiDAR with respect to time, is exported in the global frame. Second, to georeference LiDAR measurements point-wise into the global frame, each range and intensity measurement (i.e., point) is individually transformed into the global frame by using a pose of the LiDAR at the time of each measurement.

The continuous LiDAR data is split into scans, i.e., full 360° rotations of the spinning LiDAR. To compensate for motion, the LiDAR readings from each scan are reprojected into a fixed frame. The frame’s pose is defined according to the LiDAR’s pose at the time of the first scan measurement. Finally, the points associated with a scan are selected from the georeferenced point cloud and transformed into the fixed frame separately for each scan in the LiDAR data to produce final motion-compensated scans.

D. Data Recording

Overall, the dataset has 22 331 frames collected over 14.4 km of driving in three distinct scenarios, captured during both day-time and late-evening hours. Maps of the routes are provided in Fig. 3 together with sample frames from the dataset. To get diversity, we have selected and annotated a total 5 027 frames from these sequences. Fig. 3 highlights these areas on the routes where annotated images are selected. We have annotated 1 011 frames from the first scenario (*Accumulated snow* in Fig. 3), collected on December 13, 2023, in normal weather conditions with accumulated snow on the ground. The second scenario (*Falling snow*) includes 3 016 annotated frames collected on April 5, 2024, during snowfall. Finally, the third scenario (*Highway snow*) collected at late evening on December 13, 2023 includes 1 000 annotated frames. In Snowy Scenes, 3 016 frames of the total 5 027 annotated samples involve active snowfall. Similarly, around 7 K of the overall 22 K frames contain falling snowflakes.

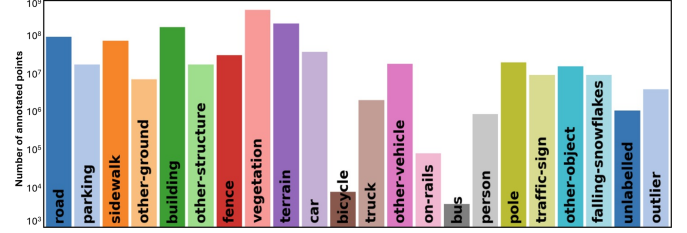


Fig. 4. Number of annotated points per class.

E. Data Privacy Protection

To ensure the privacy of all individuals in the dataset and to comply with the European General Data Protection Regulation (GDPR) [20], human faces and vehicle license plates are blurred by an open-source model from Understand AI [21].

F. Data Annotation

We provide high-quality and detailed ground-truth annotations for the full-scan LiDAR points. Skilled human annotators manually perform all annotations using the Segments.ai [22] annotation platform. All annotations then undergo rigorous quality checks. We note that detailed point-wise and bounding box annotation of LiDAR scans corrupted by snow is a highly challenging, extremely laborious, and time-consuming task because these scans contain points with indiscernible semantic content due to thick snow cover and falling snowflakes. Therefore, the annotators utilize the corresponding RGB images and Google Street views as additional references to ensure accurate point and bounding box annotations. On average, it takes about 5 hours to fully annotate a single LiDAR scan.

Two distinct annotations are available in Snowy Scenes: 3D bounding boxes and point-wise semantic labels. The former involves the annotation of each scan with precise 7 degrees of freedom bounding boxes $\{x, y, z, dx, dy, dz, \theta\}$, where (x, y, z) is the position of the center of the box, (dx, dy, dz) are respectively the length, width and height of the box and θ is its orientation.

Regarding semantic labels, all 5 027 LiDAR scans are point-wise annotated with 27 semantic classes by adopting the SemanticKITTI [14] format. We additionally provide a unique label for falling snowflakes. In our annotation set, the class *outlier* represents the power lines over the railway and undefined points. We also label points as *unlabelled* if they cannot be identified or are located below ground level due to false sensor readings. Both classes (i.e., *unlabelled* and *outlier*) are not used in our benchmarking experiments.

G. Data Statistics

Snowy Scenes has annotations of 27 semantic classes for over 5 K 3D LiDAR scans. Fig. 4 displays the distribution of annotated points for 22 classes. For visualization, in this figure, classes of moving objects are merged with their static counterparts, such as *moving car* is combined with *car* and *moving person* with *person*. These detailed statistics show that some classes, such as *bus* and *bicycle*, occur less often, whereas classes *vegetation*, *terrain*, and *building*, among others, have higher occurrence frequency. This class imbalance is common

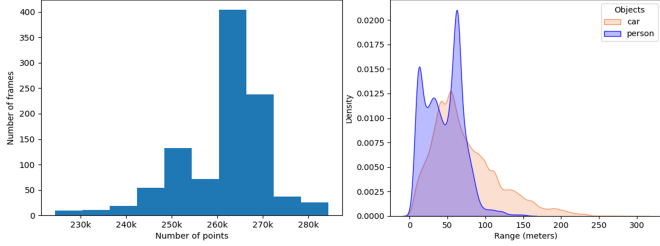


Fig. 5. Point cloud counts (left) and the object distance distribution relative to the ego vehicle (right).

in many existing datasets [10], [14] and is mainly due to the test environment with unbalanced object distribution.

A total of 221 081 box annotations are provided, mostly for cars and pedestrians, with a few trucks and on-rail boxes. Fig. 5 (left) depicts the distribution of point cloud quantity per frame. Most LiDAR point counts are concentrated between 260 K and 270 K, indicating the high point cloud density in the dataset. Fig. 5 (right) shows the distance distribution between the ego vehicle and the annotated boxes for cars and pedestrians. Car samples are dispersed over a wide range from the ego vehicle, while pedestrians are close to the ego vehicle.

H. Data Illustration

Fig. 1 illustrates a sample snowy point cloud scan captured in the late evening with ground truth bounding boxes and point-wise semantic labels, including falling snowflakes. The stitched three thermal images and the corresponding RGB image with projected points and bounding box annotations are provided at the top right.

RGB and thermal camera images are shown from five different locations around the test trajectories in Fig. 3.

Fig. 6 shows additional annotated point cloud samples from the Bird's-Eye-View (BEV) perspective with their RGB images. Falling snowflakes (cyan) are more visible in the second and third BEV projections. See more samples in the supplementary material: <https://github.com/snowyscenes/dataset>.

IV. BENCHMARK EXPERIMENTS

Snowy Scenes supports the individual and joint learning of multiple 3D perception tasks, including object detection, semantic segmentation, and point cloud denoising, in both unimodal and multimodal setups. We present a set of benchmark experiments with the corresponding metrics. The dataset was split at the scene level into 60% training (3016 scenes), 20% validation (1005 scenes), and 20% testing (1006 scenes) to ensure no scene overlap and maintain balanced class distributions across all subsets. Across all splits, the dataset maintains a consistent distribution of scene types. In each set, falling snowy scenes dominate (60%), followed by highway (20%) and accumulated snow (20%). The same splits are employed in all benchmarks, where several state-of-the-art models are evaluated to assess their performance on Snowy Scenes.

A. 3D Semantic Segmentation

We first study the 3D semantic segmentation of LiDAR point clouds. 3D semantic segmentation is a multi-class classification

task with the objective of assigning a unique class label to each point in a 3D LiDAR scan. We ignore classes *unlabeled* and *outlier* and use the remaining 25 classes of Snowy Scenes in the evaluation.

Evaluation Metrics: We assess the segmentation performance by computing the commonly applied Intersection-over-Union (IoU) metric for each class and the mean IoU (mIoU) over all classes, defined as: $IoU = \frac{TP}{TP+FP+FN}$, where TP, FP, and FN are the number of true positive, false positive, and false negative model predictions.

Baselines: Mainstream 3D semantic segmentation models rely on either point-wise, projection-based, or voxel-based point cloud representation. Therefore, we employ one state-of-the-art model from each of these approaches as baselines: point-based PTv3 [24], projection-based SalsaNext [23], and voxel-based Cylinder3D [25]. All three models are trained with 40 epochs. We use the default optimizer published with the related models.

Results: The segmentation results in Table III indicate that the voxel-based Cylinder3D [25] achieves the highest mIoU score of 76.38, surpassing PTv3 [24] (72.43) and SalsaNext [23] (66.54). This suggests that voxel-based methods are more effective for semantic segmentation in high-resolution snowy environments compared to point-based or projection-based approaches. Projection-based methods, such as range-view representations in SalsaNext [23], inherently suffer from information loss due to the many-to-one mapping problem and geometric distortions, which become more pronounced with wider fields of view and denser point clouds. While point-based methods retain fine-grained geometric details, their lower performance may stem from architectural limitations. For instance, PTv3 [24] operates with a fixed receptive field in terms of the number of neighboring points. This means that, as LiDAR resolution increases, the effective spatial coverage of this receptive field decreases. In contrast, voxel-based methods maintain a consistent spatial receptive field, regardless of LiDAR resolution, which potentially explains their superior performance.

B. 3D Point Cloud Desnowing

Point cloud desnowing, a.k.a. denoising or filtering, is the task of detecting and removing noisy points, i.e., outliers, in 3D LiDAR scans. This filtering operation is crucial to clean point clouds corrupted by inclement weather, particularly snowfall. It is because falling snowflakes can lead to a degradation in the performance of downstream semantic segmentation and object detection tasks, for instance, by forming false positives or ghost detections [26], [27], [28]. We treat the point cloud desnowing as a binary segmentation problem. Thus, each point is classified as either a falling snow particle or a regular point.

Evaluation Metrics: In addition to IoU, we evaluate the desnowing performance by computing $Precision = \frac{TP}{TP+FP}$, $Recall = \frac{TP}{TP+FN}$, and $F_1 = 2 \frac{Precision \times Recall}{Precision+Recall}$, where TP, FP, and FN define the number of true positive, false positive, and false negative predictions.

Baselines: Desnowing algorithms for LiDAR point clouds are either based on conventional statistical methods [7], [29], [31] or advanced deep learning models [26], [28], [30], [32], [33], trained in supervised or unsupervised fashions. Therefore, as baselines, we use two statistical methods, DROR [29] and SSR [28], which solely rely on distance and intensity differences between points. We further employ several advanced deep networks specifically designed to remove weather-based

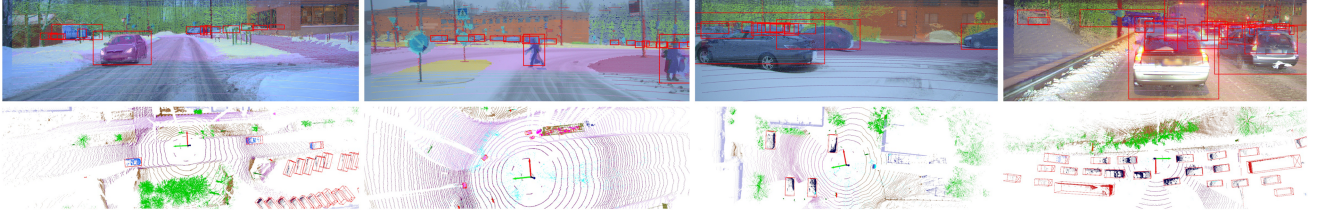


Fig. 6. Sample scenes from Snowy Scenes. The top row shows the cropped RGB images with the project 3D boxes and annotated LiDAR scans. Bird-eye-view LiDAR scans with full point-wise annotations and boxes are depicted at the bottom. Falling snowflake points are depicted in cyan color (see the second and third columns).

TABLE III
MULTI-CLASS SEMANTIC SEGMENTATION RESULTS ON THE SNOWY SCENES TEST SET

Model	car	bicycle	on-rails	truck	other-vehicle	person	road	parking	sidewalk	other-ground	building	fence	other-structure	vegetation	terrain	pole	tree	traffic-sign	other-object	moving-car	moving-person	moving-on-rails	moving-truck	moving-other-vehicle	falling-snow-flakes	mIOU
SalsaNext [23]	79.61 _{±0.30}	70.09 _{±2.84}	78.17 _{±1.85}	72.11 _{±1.84}	54.97 _{±0.80}	67.74 _{±0.82}	81.91 _{±0.09}	69.68 _{±0.13}	68.83 _{±0.22}	67.25 _{±0.42}	73.78 _{±0.11}	64.41 _{±0.13}	74.11 _{±0.02}	79.48 _{±0.08}	67.30 _{±0.08}	56.14 _{±0.24}	15.06 _{±4.05}	57.50 _{±0.22}	53.79 _{±0.34}	85.15 _{±0.06}	71.86 _{±0.43}	32.40 _{±2.14}	78.48 _{±1.06}	81.15 _{±0.54}	62.59 _{±0.19}	66.54 _{±0.30}
PTv3 [24]	84.10 _{±0.19}	31.67 _{±0.32}	76.14 _{±0.32}	75.16 _{±0.34}	61.85 _{±0.15}	70.16 _{±0.07}	77.42 _{±0.02}	60.22 _{±0.17}	61.85 _{±0.00}	51.12 _{±0.20}	95.80 _{±0.01}	85.99 _{±0.01}	92.32 _{±0.01}	93.94 _{±0.01}	74.33 _{±0.01}	80.97 _{±0.01}	31.67 _{±0.32}	81.83 _{±0.01}	74.59 _{±0.01}	71.44 _{±0.30}	83.30 _{±0.01}	47.49 _{±0.17}	78.03 _{±0.15}	84.84 _{±0.13}	88.23 _{±0.01}	72.43 _{±0.01}
Cylinder3D [25]	90.43 _{±0.20}	57.54 _{±2.80}	61.80 _{±0.26}	90.50 _{±1.19}	78.02 _{±0.57}	80.72 _{±0.44}	84.99 _{±0.32}	80.61 _{±0.51}	76.31 _{±0.45}	76.38 _{±0.35}	83.23 _{±0.13}	87.14 _{±0.12}	92.17 _{±0.19}	89.34 _{±0.01}	79.64 _{±0.27}	72.11 _{±0.21}	17.40 _{±1.32}	77.20 _{±0.26}	77.39 _{±0.21}	86.05 _{±0.11}	80.25 _{±0.06}	24.45 _{±1.15}	90.79 _{±1.47}	91.29 _{±0.36}	83.74 _{±0.47}	76.38 _{±0.14}

TABLE IV
DE-SNOWING (BINARY SEGMENTATION) RESULTS ON THE SNOWY SCENES TEST SET. SL AND UN STAND FOR SUPERVISED AND UNSUPERVISED LEARNING, RESPECTIVELY. HIGHER (↑) IS BETTER

Model	Type	Precision↑	Recall↑	F ₁ ↑	IOU↑
SalsaNext [23]	SL	74.99 _{±0.46}	86.93 _{±0.12}	80.52 _{±0.32}	67.39 _{±0.44}
Cylinder3D [25]	SL	95.87 _{±0.55}	88.20 _{±0.63}	91.87 _{±0.14}	84.96 _{±0.23}
3D-OutDet [26]	SL	90.70 _{±0.23}	83.24 _{±0.03}	86.81 _{±0.09}	76.70 _{±0.14}
DROR [29]	UN	6.52	81.84	12.08	6.43
SSR [28]	UN	40.23	74.79	52.32	35.43
LiSnowNet-L1 [30]	UN	12.34 _{±0.86}	79.76 _{±5.91}	21.25 _{±1.08}	11.90 _{±0.67}
LiSnowNet-L2 [30]	UN	13.72 _{±0.00}	6.66 _{±0.00}	8.96 _{±0.00}	4.69 _{±0.00}
3D-UnOutDet [28]	UN	49.82 _{±1.29}	82.65 _{±0.09}	62.14 _{±0.97}	45.09 _{±1.03}

noise (e.g., snowflakes or raindrops) from LiDAR point clouds. For instance, LiSnowNet [30] detects snow particles based on the number of Fourier or Wavelet coefficients of point clouds in an unsupervised manner. 3D-OutDet [26] operates directly on the point level by applying neighborhood convolution to explore noisy nearest neighbors only. 3D-UnOutdet [28] is the unsupervised counterpart of 3D-OutDet [26]. Finally, recalling that desnowing is a binary classification task, we also investigate the performance of general-purpose 3D semantic segmentation models such as SalsaNext [23] and Cylinder3D [25] on this filtering task.

Results: The benchmark results are summarized in Table IV. We categorize the evaluated models based on whether they operate in a supervised or unsupervised setting. Among the supervised methods, the voxel-based model Cylinder3D [25] achieves the highest scores in all metrics. The same reasoning applies to the point-based model 3D-OutDet [26], which makes it inefficient for high-resolution LiDAR data. In contrast, Cylinder3D [25] is less dependent on the LiDAR resolution, while still preserving fine-grained details through its voxelization strategy.

Unexpectedly, all unsupervised methods underperform in contrast to their supervised counterparts. LiSnowNet [30] demonstrates poor performance, likely due to its architectural design, which is tailored to 32- and 64-beam LiDAR data and does not transfer effectively to the higher-resolution 128-beam

LiDAR in Snowy Scenes. These findings clearly show that unsupervised state-of-the-art point cloud denoising methods are not resolution-agnostic and require further study.

C. 3D Object Detection

3D object detection is the task of assigning an accurate 3D bounding box around an object. The process involves LiDAR and/or RGB camera data and consists of detecting and classifying objects, such as cars and pedestrians, in the surrounding scene. In adverse weather, such as snow, 3D object detectors may suffer from ghost detections and false positives due to noise in point clouds or camera images. For this benchmark experiment, we only focus on the *Car* class.

Evaluation Metrics: We evaluate the object detection task following the protocol established by nuScenes [10]. A predicted bounding box is considered a true positive if its distance to the corresponding ground truth box is below a specified threshold. The average precision (AP) is then computed as $AP = \int_0^1 p(r) dr$, where $p: r \rightarrow p(r)$ denotes the precision-recall curve. The integral is approximated using the standard 40 recall points method. For evaluation, we consider four distance thresholds: 0.5 m, 1 m, 2 m, and 4 m.

In addition to AP, we report several complementary metrics: the Average Translation Error (ATE), which quantifies the positional error in the Bird's-Eye-View (BEV); the Average Scale Error (ASE), which measures discrepancies in object size between predictions and ground truth; and the Average Orientation Error (AOE), which captures differences in orientation angles. The final AP score is obtained by averaging the AP values across the four distance thresholds. For the other metrics (ATE, ASE, and AOE), we report results computed using a distance threshold of 2 m.

Baselines. For benchmarking, we implement established methods on Snowy Scenes using the OpenPCDet [34] codebase. We train all models for 80 epochs with AdamW and a learning rate of 0.003. We evaluate three categories of LiDAR-based object detection: BEV-based (PointPillars [35], CenterPoint [36], Transfusion-L [37]), voxel-based (VoxelNext [38]), and point-based (IA-SSD [39]). To assess multimodal detection, we apply

TABLE V

CAR 3D OBJECT DETECTION PERFORMANCE ON THE SNOWY SCENES TEST SET

Method	Modality	AP $\uparrow$	ATE $\downarrow$	ASE $\downarrow$	AOE $\downarrow$
PointPillars [35]	L	75.2	0.24	0.063	0.280
CenterPoint [36]	L	66.9	0.24	0.072	0.414
TransFusion-L [37]	L	76.2	0.23	0.097	0.391
VoxelNeXt [38]	L	74.1	0.18	0.064	0.530
IA-SSD [39]	L	72.1	0.20	0.051	0.369
Center Point (FOV) [39]	L	35.0	0.62	0.071	0.445
TransBEVFusion	L	67.63	0.37	0.11	1.65
TransBEVFusion	LC	77.77	0.28	0.12	1.65
TransBEVFusion	LCT	78.52	0.28	0.12	1.66
UniTR [40]	L	51.23	0.36	0.11	0.14
UniTR [40]	LC	71.74	0.30	0.10	0.89
UniTR [40]	LCT	70.65	0.30	0.11	0.91

UniTR [40] and our custom fusion model TransBEVFusion, which is based on the BEVFusion model [41].

Results: Table V summarizes the benchmark results. TransFusion [37] outperforms VoxelNeXt [38] by 2.1 AP. This indicates that, unlike the trends observed in semantic segmentation and point cloud denoising benchmarks, BEV-based detectors achieve better performance in snowy conditions compared to voxel- and point-based approaches. However, the performance gap relative to voxel-based detectors remains modest. Point-based models, such as IA-SSD [39], obtain the lowest AP due to limited spatial coverage within their receptive fields. These findings suggest that state-of-the-art point-based detectors trained on low-resolution, noise-free LiDAR data may struggle to generalize to the noisy, high-resolution environments characteristic of Snowy Scenes.

Additionally, applying our custom multimodal detector TransBEVFusion and the original UniTR [40] yields a significant performance improvement compared to their LiDAR-only counterparts, showing the importance of sensor fusion. Detailed results and further analysis of these fusion models are provided in the supplementary material.

D. Multitask Learning

Recent advances in perception have demonstrated that leveraging multiple tasks can improve performance on individual tasks. In the context of 3D perception, models such as Lidar-MultiNet [42] and LidarFormer [43] use both object detection and semantic segmentation supervision to enhance the accuracy of each task. Our dataset, with its comprehensive set of annotations, enables models to be trained in a multitask setting. In this respect, we implement a baseline multitask model by extending the OpenPCDet framework [34] to support the joint learning of both semantic segmentation and object detection tasks using LiDAR point clouds only. The objective of this experiment is not to achieve state-of-the-art performance, which is beyond the scope of this work. Rather, it is to illustrate the potential of the Snowy Scenes dataset and to contribute to the community through this new multitask framework.

Evaluation Metrics: We assess segmentation and detection performance by computing mIoU and AP scores, respectively.

Baselines: Due to the lack of source code for Lidar-MultiNet [42] and LidarFormer [43], we implement a naive multitask baseline model, named Snowy Scenes MultiTask (SSMT). SSMT employs the voxel-based semantic segmentation model MinkUNet [44] as a backbone and combines it with a BEV-based feature extractor followed by the detection head

TABLE VI

MULTITASK LEARNING RESULTS ON THE SNOWY SCENES TEST SET. $\uparrow$ DENOTES HIGHER IS BETTER

Model	mIoU $\uparrow$	Car AP $\uparrow$
SSMT-seg (ours)	53.4	-
SSMT-det (ours)	-	64.5
SSMT-joint (ours)	48.3	67.2

CenterPoint [36]. The SSMT model is trained by fine-tuning the joint model over 10 epochs with a learning rate of 1e-4, after pretraining each branch individually. Since the joint model shares the backbone weights, we keep the segmentor's weights.

Results: The obtained results are presented in Table VI. The multitask model SSMT-joint outperforms its single-task counterparts in detection, but underperforms in semantic segmentation. This outcome highlights the need for further optimization in the training procedure. We believe that the increase in detection performance is due to the semantic signal, which helps correctly regress the car boxes with rich context information. However, the detector signal is redundant for most semantic classes. Nevertheless, the reduction in segmentation performance is primarily due to the challenging and underrepresented classes, as the most frequent classes retain high accuracy.

V. DISCUSSION & CONCLUSION

This letter introduces Snowy Scenes, the first multimodal dataset collected in real-world snowy conditions using high-resolution 128-beam LiDAR, RGB, and thermal cameras coupled with GNSS/IMU. Leveraging our 5 K rich annotated data, our extensive benchmarking experiments with state-of-the-art models show that voxel-based methods excel in semantic segmentation and noise removal tasks. Their capacity to capture fine-grained information while being agnostic to LiDAR resolution makes them the optimal choice for off-the-shelf adaptation. When it comes to the object detection task, BEV-based methods outperform other approaches. Across all three benchmarking tasks, point-based models exhibit low performance due to their limited receptive fields. Additionally, our results reveal that unsupervised point cloud denoising methods are not resolution-agnostic and require further study. Additional experiment results are provided in the supplementary material available at <https://github.com/snowyscenes/dataset>.

As future work, leveraging GNSS/IMU data in Snowy Scenes, we plan to conduct additional experiments to benchmark SLAM algorithms under snowy conditions and address potential challenges.

Limitations: Snowy Scenes has no Radar sensor. The dataset contains only front-view limited RGB and thermal cameras. There is no annotation of RGB or thermal images. No other weather conditions, such as rain and fog, are included. Localization- and mapping-related benchmarks are excluded.

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